

RBF-FD Solver: Least Squares Method for Operator Reproduction

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1 Problem Setting:

Following the standard RBF-FD methods, let $u \in \mathcal{F}$ be a solution for a PDE in form of:

$$-Lu = -\nabla \cdot (A \nabla u) = f \quad \text{in } \Omega, \quad (1)$$

Let $1 < p < \infty$, $A \in C^{0,1}(\bar{\Omega}; \mathbb{R}^{d \times d}) : \bar{\Omega} \rightarrow \mathbb{R}^{d \times d}$ is an symmetric positive definite anisotropic tensor, where

$$\eta |\xi|^2 \leq \mathbf{A}(\mathbf{x}) \xi \cdot \xi \leq \mathbf{H} |\xi|^2, \quad \mathbf{0} < \eta \leq \mathbf{H} \quad (2)$$

and $f \in L^p(\Omega)$ is the forcing term acting in the domain Ω . The problem is subject to Neumann and Dirichlet Boundary Conditions partitioned within $\partial\Omega$ of the form:

$$u = g_D \quad \text{on } \Gamma_D \subset \partial\Omega, \quad (3)$$

$$\frac{\partial u}{\partial n} \equiv n \cdot A \nabla u = g_N \quad \text{on } \Gamma_N \subset \partial\Omega, \quad (4)$$

For simplicity, we assume $\partial\Omega = \Gamma_D \cup \Gamma_N$ and $\Gamma_D \cap \Gamma_N = \emptyset$. Furthermore, let $g_D, g_N \in W^{k-1,p}(\partial\Omega)$. By *Agmon-Douglis-Nirenberg* Theorem, we have the following estimate for u :

$$\|u\|_{W^{k,p}(\Omega)} \leq C \left(\|Lu\|_{L^p(\Omega)} \sum_{j=1}^m \|g_j\|_{W^{k-1,p}(\partial\Omega)} + \|u\|_{L^p(\Omega)} \right). \quad (5)$$

2 RBF-FD Method:

2.1 Local Approximation setup

To obtain an approximation of the operator L at point $\mathbf{z}$, we can approximate $\hat{u} \approx u$ on a neighborhood $\Omega \subseteq \mathbb{R}^n$ of $\mathbf{z}$ such that we approximate $Lu \approx L\hat{u}$. We define the radial basis functions in the form $\phi_i(\mathbf{x}) := \phi(\|\mathbf{x} - \mathbf{y}_i\|) : \Omega \rightarrow \mathbb{R}$ for $i = 1, \dots, m$ and centers $\{\mathbf{y}_i\}_{i=1}^m \subseteq \Omega$. Define the vector space $V_Y = \text{span}\{\phi_1, \dots, \phi_m\}$. Let $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \Omega$ be a set of stencil nodes for a point $\mathbf{z}$ in Ω . For the classical approximation scheme, let $X = Y$ and $m = n$. We can now define $\Phi \in \mathbb{R}^{n \times m}$, where $\Phi_{ij} = \phi_j(\mathbf{x}_i)$. The role of these basis functions will be to approximate the operator L . The approximation of u takes the form of:

$$\hat{u}(\mathbf{x}) = \sum_{i=1}^n \lambda_i \phi_i(\mathbf{x}) \quad (6)$$

This will lead to the form $\Phi \lambda = \hat{\mathbf{u}}$ for $\hat{\mathbf{u}} = [u(\mathbf{x}_1), \dots, u(\mathbf{x}_n)]^T$. We can simply for λ to obtain:

$$\hat{u}(\mathbf{x}) = \sum_{i=1}^n (\Phi^{-1} \mathbf{u})_i \phi_i(\mathbf{x}) = \phi(\mathbf{x})^T \Phi^{-1} \mathbf{u} \quad (7)$$

The approximation of Lu at z is in the form of:

$$Lu(\mathbf{z}) \approx L\hat{\mathbf{u}}(\mathbf{z}) = \mathbf{L} \phi^T(\mathbf{z}) \lambda = \mathbf{L} \phi^T(\mathbf{z}) \Phi^{-1} \mathbf{u} := \mathbf{w}^{\mathbf{L}, \mathbf{X}}(\mathbf{z}) \mathbf{u} \quad (8)$$

Thus, approximation the operator turns into local solver for the weights w_i from the form:

$$\Phi^T \mathbf{w}^{\mathbf{L}, \mathbf{X}}(\mathbf{z}) = \mathbf{L} \phi(\mathbf{z}) \quad (9)$$

Through the method of undefined coefficients on the basis $\{\phi_i\}$ we get the same resulting set of weights. Finally, by defining the basis for V_Y as radial basis functions results in the Radial Basis Function generated Finite Difference method (RBF-FD).

2.2 Polynomial Augmentation

For conditionally positive definite kernels, we augment them through the form:

3 Weight Constraint RBF-FD Setup:

Rather than assuming a form for the solution, our goal is to approximate the operator through weights dual to u sampled at nodes. Suppose $m > n$, thus more center nodes than stencil nodes per stencil. Additionally, instead of assuming exactness of the form $L\phi_j(\mathbf{x}) = \sum_{i=1}^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i)$ for $j = 1, \dots, m$ as in the classical RBF-FD, we define a residual function:

$$r_j(\mathbf{x}) := L\phi_j(\mathbf{x}) - \sum_{i=1}^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i) \quad (10)$$

$$r(\mathbf{x}) = L\phi(\mathbf{x}) - \Phi^T \mathbf{w}(\mathbf{x}), \quad \mathbf{w} : \mathbb{R}^n \rightarrow \mathbb{R}^n \text{ for } \mathbf{x} \in \Omega. \quad (11)$$

Suppose $\{p_i\}_{i=1}^l$ is a set of polynomials augmenting the radial basis functions. Let $\Pi_k = \text{span}\{p_i\}_{i=1}^l$ with polynomial degree up to k . We want to reproduce the operator exactly on the polynomial basis:

$$Lp_j(\mathbf{x}) := \sum_{i=1}^n w_i(\mathbf{x})p_j(\mathbf{x}_i) \text{ for } j = 1, \dots, l \quad (12)$$

$$L\mathbf{p}(\mathbf{x}) = \mathbf{P}^T \mathbf{w} \quad (13)$$

where $\mathbf{P}_{ij} := p_j(\mathbf{x}_i) \in \mathbb{R}^{n \times l}$. Thus, we have a constrained least squares problem in the form:

$$\begin{cases} \min J(\mathbf{w}(\mathbf{x})) := \min \frac{1}{2} \|\mathbf{r}(\mathbf{x})\|_2^2 & \text{for } \mathbf{x} \in \Omega \\ \text{subject to } \Phi^T \mathbf{w} = L\mathbf{p} \end{cases} \quad (14)$$

We can solve this setup by defining the Lagrangian $\mathcal{L}$ for the problem as follows:

$$\begin{aligned} \mathcal{L}(\mathbf{w}, \boldsymbol{\nu}) &:= J(\mathbf{w}) + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - L\mathbf{p}(\mathbf{x}) \rangle \\ &= \frac{1}{2} \langle \Phi^T \mathbf{w}, \Phi^T \mathbf{w} \rangle - \langle \Phi^T \mathbf{w}, L\phi \rangle + \frac{1}{2} \langle L\phi, L\phi \rangle + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - L\mathbf{p} \rangle \end{aligned}$$

4 Consistency Conditions for Finite Spaces and Interpolation Results

The resulting KKT equations for this system are then:

$$\nabla_{\mathbf{w}} \mathcal{L} = 0 : \Phi L\phi = \Phi \Phi^T \mathbf{w} + \mathbf{P}\boldsymbol{\nu} \quad (15)$$

$$\nabla_{\boldsymbol{\nu}} \mathcal{L} = 0 : \mathbf{P}^T \mathbf{w} = L\mathbf{p} \quad (16)$$

The augmented matrix then takes the form:

$$\begin{bmatrix} \Phi \Phi^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{w} \\ \boldsymbol{\nu} \end{bmatrix} = \begin{bmatrix} \Phi L\phi \\ L\mathbf{p} \end{bmatrix} \quad (17)$$

The primary goal is to approximate $Lu(\mathbf{x}) \approx \mathbf{w}^T(\mathbf{x})\mathbf{u}$, with $\mathbf{u} = [u(\mathbf{x}_1), \dots, u(\mathbf{x}_n)]^T$. Define $L_h f(\mathbf{x}) := \mathbf{w}^T(\mathbf{x})\mathbf{f}$ for some function f . Thus, we have the following:

$$\begin{aligned}
L_h f(\mathbf{x}) &= \mathbf{w}^T(\mathbf{x})\mathbf{f} \\
&= [\mathbf{w}^T \quad \boldsymbol{\nu}^T] \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \\
&= \left(\begin{bmatrix} \boldsymbol{\Phi}\boldsymbol{\Phi}^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \boldsymbol{\Phi}L\phi \\ L\mathbf{p} \end{bmatrix} \right)^T \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \\
&= [L\phi^T \boldsymbol{\Phi}^T \quad L\mathbf{p}^T] \begin{bmatrix} \boldsymbol{\Phi}\boldsymbol{\Phi}^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix}^{-T} \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \\
&= [L\phi^T \boldsymbol{\Phi}^T \quad L\mathbf{p}^T] \begin{bmatrix} \mathbf{q} \\ \boldsymbol{\alpha} \end{bmatrix} \\
&= L\phi^T(\mathbf{x}) (\boldsymbol{\Phi}^T \mathbf{q}) + L\mathbf{p}^T(\mathbf{x})\boldsymbol{\alpha}
\end{aligned}$$

Next, we can show that the coefficients $\mathbf{q}, \boldsymbol{\alpha}$ relate to the coefficients of u as follows:

$$\begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi}\boldsymbol{\Phi}^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \boldsymbol{\alpha} \end{bmatrix} \quad (18)$$

Define $V_h = \{\rho(x) : \rho(x) = \sum_{i=1}^m \lambda_i \phi_i(x) \text{ such that } \lambda \in \text{Range}(\boldsymbol{\Phi}^T)\} \subseteq V_Y$. Let $s \in V_h + \Pi_k$. Thus, $s(x) = \phi^T(\mathbf{x})\boldsymbol{\Phi}^T \mathbf{q} + \mathbf{p}^T(\mathbf{x})\boldsymbol{\alpha}$ where $\mathbf{q} \in \mathbb{R}^n$ and $\boldsymbol{\alpha} \in \mathbb{R}^l$. Then:

$$s = s(X) = \boldsymbol{\Phi}\boldsymbol{\Phi}^T \mathbf{q} + \mathbf{P}\boldsymbol{\alpha} = f(X) = \mathbf{f} \quad (19)$$

$$\mathbf{P}^T \mathbf{q} = \mathbf{0} \quad (20)$$

Hence, we get interpolation of f at the node set X . Furthermore, from the KKT matrix we can show that $\boldsymbol{\Phi}\mathbf{r} = \mathbf{P}\boldsymbol{\nu}$ which implies that $\boldsymbol{\Phi}\mathbf{r} \in \text{range}(\mathbf{P}) = (\ker(\mathbf{P}^T))^\perp$. Thus, we get the relationship:

$$\mathbf{r}^T (\boldsymbol{\Phi}^T \mathbf{z}) = \mathbf{0} \text{ for } \mathbf{P}^T \mathbf{z} = \mathbf{0} \quad (21)$$

Finally, we check the consistency error for $s \in V_h + \Pi_k$ with $\mathbf{P}^T \mathbf{q} = \mathbf{0}$.

$$\begin{aligned}
Ls(x) - L_h s(x) &= L(\phi^T(\mathbf{x})\boldsymbol{\Phi}^T \mathbf{q} + \mathbf{p}^T(\mathbf{x})\boldsymbol{\alpha}) - \mathbf{w}^T(\mathbf{x})s \\
&= L\phi^T(\mathbf{x})\boldsymbol{\Phi}^T \mathbf{q} + L\mathbf{p}^T(\mathbf{x})\boldsymbol{\alpha} - \mathbf{w}^T(\mathbf{x})(\boldsymbol{\Phi}\boldsymbol{\Phi}^T \mathbf{q} + \mathbf{P}\boldsymbol{\alpha}) \\
&= L\phi^T(\mathbf{x})\boldsymbol{\Phi}^T \mathbf{q} + L\mathbf{p}^T(\mathbf{x})\boldsymbol{\alpha} - \mathbf{w}^T(\mathbf{x})\boldsymbol{\Phi}\boldsymbol{\Phi}^T \mathbf{q} - \mathbf{w}^T(\mathbf{x})\mathbf{P}\boldsymbol{\alpha} \\
&= (L\phi(\mathbf{x}) - \boldsymbol{\Phi}^T \mathbf{w}(\mathbf{x}))^T \boldsymbol{\Phi}^T \mathbf{q} + (L\mathbf{p}^T(\mathbf{x}) - \mathbf{P}^T \mathbf{w}(\mathbf{x}))^T \boldsymbol{\alpha} \\
&= \mathbf{r}^T \boldsymbol{\Phi}^T \mathbf{q} + (L\mathbf{p}(\mathbf{x}) - \mathbf{P}^T \mathbf{w}(\mathbf{x}))^T \boldsymbol{\alpha}
\end{aligned}$$

$$Ls(x) - L_h s(x) = 0 \text{ due to (16) and (21)} \quad (22)$$

5 Consistency Results for Function Approximations

Let u be a solution to (1) and $s \in V_h + \Pi_k$. We can say $u = s + (u - s)$ to get the following:

$$\begin{aligned}
\mathcal{E}[u](\mathbf{x}) &:= Lu(\mathbf{x}) - L_h u(\mathbf{x}) \\
&= Lu(\mathbf{x}) - \mathbf{w}^T(\mathbf{x})\mathbf{u} \\
&= L((u - s) + s)(\mathbf{x}) - \mathbf{w}^T(\mathbf{x})\mathbf{u} - L_h s(\mathbf{x}) + L_h s(\mathbf{x}) \\
&= L(u - s)(\mathbf{x}) + (Ls(\mathbf{x}) - L_h s(\mathbf{x})) + L_h(s - u)(\mathbf{x}) \\
&= L(u - s)(\mathbf{x}) - \mathbf{w}^T(\mathbf{x})(\mathbf{u} - \mathbf{s}), \text{ by (22)} \\
&= Le(\mathbf{x}) - \mathbf{w}^T(\mathbf{x})\mathbf{e}, \mathbf{e}(\mathbf{x}) = u(\mathbf{x}) - s(\mathbf{x}) \\
&= \mathcal{E}[\mathbf{e}](\mathbf{x})
\end{aligned}$$

References

- [1] Agmon, S., Douglis, A. and Nirenberg, L. (1959). *Estimates near the boundary for solutions of elliptic partial differential equations satisfying general boundary conditions*. I. Comm. Pure Appl. Math., 12: 623-727.
<https://doi.org/10.1002/cpa.3160120405>
- [2] Slak, J. (2026). *Adaptive RBF-FD method*. PhD Thesis, University of Ljubljana.